NORTH SOUTH UNIVERSITY
Department of Electrical and Computer Engineering

CSE 115L — Programming Language I Lab

LABORATORY TASK SHEET

Lab No. **09** | Repetition — Counting Loops, `while` and `for`

Course faculty: **Mirza Mohammad Lutfe Elahi** | Lab instructor: **Moshiur Rahan** | Fall 2026

Student name		**Student ID**	
Section		**Date**	
Team / Group		**Worksheet**	Ch 5, §5.1–§5.4

MARK DISTRIBUTION *Lab performance, out of 10, awarded in the room.*

Task	Description	Tier	Marks	Awarded
E9.1	Count-down `while` loop; the count **and** the control variable's value after the loop both printed from variables	Easy	1	
E9.2	The same loop as a single `for`; output byte-for-byte identical to E9.1, both files shown together	Easy	1	
M9.1	Sum, product and average in one pass, each accumulator correctly initialised, compound assignment used; the `0` input explained	Mediocre	2	
M9.2	Three-column table of eight multiples of **N**, aligned with `%nd` / `%n.mf` for *any* four-digit N, tested with `9999`	Mediocre	1	
M9.3	The off-by-one found, the one-character fix made, and the wrong loop **job** named	Mediocre	1	
C9.1	Powers of two to 40 from an `int` accumulator; first wrong power identified and explained from the size of an `int`; `double` trade-off stated	Challenging	2	
C9.2	All four defects found and fixed one at a time, each sorted into *warned* or *nothing*, and both follow-up answers given	Challenging	2	
		TOTAL	**10**	

Lab instructor's remarks	

Student's signature _____

Date: _____

Lab instructor's signature _____

Moshiur Rahan

Rules. (1) Nothing is marked after you leave the room — no e-mailed submissions. **(2)** Work must run on the machine in front of you; a program that does not compile scores zero for that task. **(3)** Tasks using **N** must use *your own* ID — **N is digits 4 to 7** — so two students with identical output have copied. **(4)** Guided Tasks G9.1 and G9.2 carry no marks, but the exercises above assume you have done them. **(5)** `-Wall` must be switched on; two of C9.2's four defects are reported only with it, and the instructor will check the build log. **(6) No `do-while`, no nested loops and no arrays** — every task here is solvable with `while` or `for` alone, and using next week's material scores no extra marks. **(7) Lab Quiz 2** is at the *start of next week's lab*: 40 minutes, closed book, Labs 05–08. **(8)** This signed sheet is your record of the marks; keep it until the lab grades are published.